

Petar Mircheski

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SUMMARY

Software engineer with 6+ years of experience in ML engineering, full-stack development, and high-performance scientific computing. Built production AI systems for architectural analysis at a startup; designed a simulation framework processing millions of computations per run on 1000+ node networks at Institute of Science Tokyo. Miura Award recipient (top 220 graduate engineering students in Japan, JSME); MEXT scholar; 3 peer-reviewed publications in Chaos (American Institute of Physics) and Springer; 8 international conference presentations. Completing a funded PhD (exp. April 2027); actively seeking full-time software engineering roles in Tokyo. Conversational Japanese.

EXPERIENCE

Researcher and Engineer (Funded Position)

Apr. 2022 – Present

Institute of Science Tokyo (formerly Tokyo Institute of Technology)

Tokyo, Japan

- Built a high-performance Python simulation framework.
- Processes millions of computations per run on complex network models of 1000+ nodes.
- Achieved order-of-magnitude speed-ups via JIT-compiled solvers (Numba, JAX).
- Engineered reproducible experiment pipelines with automated data storage and containerised environments.
- 3 peer-reviewed papers in Chaos (American Institute of Physics) and Springer; 8 international conference presentations.
- *Stack: Python, Numba, NumPy, SciPy, JAX, Docker, Git, Linux, Julia*

Software Engineer — Computer Vision & Full-Stack

Jun. 2019 – Aug. 2023

Eriden LLC.

Skopje, North Macedonia

- Built production AI pipelines for architectural floor-plan analysis.
- Worked on ML classification, image segmentation, 3D visualisation.
- Implemented CV algorithms for floor-plan vectorisation and spatial reconstruction.
- Shipped TypeScript/React geometry algorithms for real-time polygon rendering at 60 fps.
- Built scraping and ingestion infrastructure, cutting manual preprocessing time by 50%.
- *Stack: Python, TypeScript, React, PyTorch, OpenCV*

EDUCATION

Institute of Science Tokyo (formerly Tokyo Institute of Technology)

Tokyo, Japan

PhD, Systems & Control Engineering

Apr. 2024 – Apr. 2027 (exp.)

- Research in nonlinear dynamics and network science.
- MEXT scholarship recipient.

Tokyo Institute of Technology

Tokyo, Japan

MSc, Systems & Control Engineering

Apr. 2022 – Apr. 2024

- GPA: 3.58 / 4.0.
- Miura Award — top 220 graduate engineering students in Japan (JSME).
- Thesis: “Phase-amplitude reduction and optimal phase locking of collectively oscillating networks.”

Ss Cyril and Methodius University, Faculty of Electrical Engineering and IT

Skopje, North Macedonia

BSc, Computer Systems Engineering, Automation and Robotics

Sep. 2016 – Sep. 2020

- GPA: 9.07 / 10.
- Ministry of Education scholarship for academic excellence.
- Thesis: “Non-linear Analysis of Neural Interactions.”

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, Julia, SQL, MATLAB, LaTeX

ML / AI: PyTorch, JAX, Scikit-learn, NumPy, Numba, SciPy, Pandas, OpenCV

Front-End: React, Next.js, Tailwind CSS, HTML5/CSS, Material UI

Back-End: FastAPI, Django, Flask, REST API, PostgreSQL

Tools: Docker, Linux/Bash, Git, GitHub Actions

Spoken: English (fluent), Japanese (conversational), Macedonian (native), Serbo-Croatian (fluent)

PROJECTS

Personal Portfolio & Research Website | petarmircheski.com

2024 – Present

- Designed and deployed a full-stack personal website.
- Showcasing professional profile, research, publications, and hobbies.
- *Stack: React, Next.js, Tailwind CSS*

AWARDS & SCHOLARSHIPS

Miura Award <i>Top 220 graduate engineering students in Japan, outstanding research contributions (JSME)</i>	2024
MEXT Scholarship <i>Competitive national award for foreign researchers; granted by the Japanese Embassy</i>	2021 – Present
Ministry of Education Scholarship <i>National award for academic excellence (undergraduate)</i>	2017 – 2020
ICPC SEERC 3rd Place <i>South-Eastern European Regional Contest (1st and 3rd year of undergraduate)</i>	2017, 2019

SELECTED PUBLICATIONS

- **Mircheski, Nakao.** “Spatial locking of chimera states to frequency heterogeneity in nonlocally coupled oscillators.” *Chaos* 35, 063105, 2025.
- **Mircheski, Zhu, Nakao.** “Phase-amplitude reduction and optimal phase locking of collectively oscillating networks.” *Chaos* 33, 103111, 2023.
- **Mircheski, Zhu, Nakao.** “Phase-amplitude reduction of limit-cycling networks for optimal synchronization.” *IUTAM Bookseries* vol. 43, Springer, 2025.